

MICHAEL BAGNOLI

Chief Technology Officer (CTO) | Senior Enterprise AI Architect & Tech Entrepreneur

Location: Pisa, Italy • Mobility: Full-Remote / Hybrid / Relocation

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Work Authorization: EU Citizen

PROFILE & BACKGROUND SUMMARY

Education & Academic Background

- **B.Sc. in Statistics (Corso di Laurea Triennale in Statistica)** (*In Progress / Ongoing*)
- *Core Focus:* Statistical Modeling, Probability Theory, Predictive Analytics, Regression Analysis, Quantitative Methods, Data Architecture.
- **Liceo Scientifico Ulisse Dini (Pisa) – EsaBac Double Diploma**
- *Qualifications:* Italian Maturità Scientifica + French Baccalauréat (Double Italian-French Scientific Diploma).

Languages & Communication

- **Italian:** Native Speaker
- **English:** C1 – Advanced / Professional Working Proficiency
- **French:** C1 – Advanced / Professional Working Proficiency (EsaBac Certified)

Key Ventures & Entrepreneurial Projects (Founded)

- **ForMinds** – *Co-Founder & Chief Technology Officer (CTO)*
 - Enterprise AI consulting, corporate training, and proprietary pharmaceutical demand forecasting platform.
 - **Ultralimes Marketing** – *Founder & Lead Software Architect*
 - Proprietary market intelligence SaaS, custom predictive algorithms, and luxury multi-property PMS platforms.
 - **Golden Bloom Studio** – *Co-Owner & Head of Video Production*
 - High-end video production, digital intermediate workflows, and multimedia asset pipelines for luxury brands and global industrial leaders (Cantieri Navali Codecasa, Solvay Group).
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Passions & Extracurricular Activities

- **Writing & Literature:** Creative and technical storytelling, analytical synthesis, and clear communication of complex systems.
- **Opera & Classical Singing:** Vocal discipline, performance psychology, stage presence, and executive public speaking.
- **Track & Field / Athletics:** Physical discipline, endurance, grit, and continuous goal-driven self-improvement.
- **Photography & Visual Arts:** Visual composition, lighting design, color grading, and advanced digital imaging.

PROFESSIONAL SUMMARY

Chief Technology Officer (CTO), Tech Entrepreneur, and Top-Tier Enterprise AI Integration Consultant, featuring a polyhedric background that bridges the analytical rigor of **Quantitative Statistics** with refined **creative and artistic direction** (digital cinematography, photography, and storytelling).

Throughout my advisory and masterclass engagements with premier industrial and institutional leaders (**European Space Agency - ESA, EY, UniCredit, Banca Monte dei Paschi di Siena, Autostrade per l'Italia, Piaggio Group, Lear Corporation, Triboo Group**), my mandate has never been delivering abstract, generic lectures on AI, but **surgically embedding artificial intelligence into core operational workflows**. This continuous, hands-on immersion has given me firsthand mastery of corporate governance, strict regulatory compliance, enterprise security, and multi-scale organizational dynamics across multinational giants.

This foundation powers my entrepreneurial ventures: as **Co-Founder & CTO of ForMinds**, I lead the development of an ultra-precise, GxP-compliant **Pharmaceutical Demand Forecasting** platform; as **Founder of Ultralimes Marketing**, I designed, built, and deployed a cloud SaaS market intelligence platform with proprietary predictive algorithms on Linux infrastructure. I command the full frontier of modern innovation: from **Agentic AI** systems (CrewAI, LangChain, n8n) and evaluation frameworks (DeepEval, MLflow) to multimodal generative vision (Stable Diffusion, ComfyUI, Google Gemini), cloud-native MLOps (AWS SageMaker, Docker), and modern full-stack engineering (Python/FastAPI, Java/Spring Boot, React/Three.js/GSAP).

WORK EXPERIENCE

Co-Founder & Chief Technology Officer (CTO) | Head of AI Solutions | ForMinds

2025 – Present • Italy (Remote / Hybrid)
(Conceived in 2025 • Officially Operating in 2026 with Mirko Campari (Manager), Edoardo Del Sarto, Tommaso Muraca)

- **Executive Technical Leadership & Company Foundation:** Co-founded **ForMinds** alongside Mirko Campari (Manager), Edoardo Del Sarto, and Tommaso Muraca, spearheading the company's technical vision, engineering standards, custom software development, and enterprise AI consulting roadmap.
- **Tier-1 Enterprise Delivery Vehicle:** Operationalized ForMinds as the strategic delivery vehicle for high-stakes AI integration, custom software architectures, and executive prompt engineering for major corporate accounts, including **UniCredit, Banca Monte dei Paschi di Siena (MPS), Autostrade per l'Italia, Piaggio Group, and Lear Corporation**.
- **Flagship Pharmaceutical Forecasting Engine (R&D & Production):** Spearheading the architecture and engineering of a proprietary, end-to-end **Pharmaceutical Demand & Sales Forecasting Platform**; overseeing the full pipeline from quantitative mathematical modeling to microservice API packaging and cloud deployment.
- **Mission-Critical Precision, Speed & Security:** Architected the pharma engine for ultra-high mathematical precision, sub-second inference latency, strict data governance, end-to-end encryption, and compliance with rigorous pharmaceutical industry standards (**GxP / GDPR**).
- **Corporate IT & AI Academies:** Designed and directed bespoke enterprise academy tracks in modern software architecture, applied generative AI, and agentic workflows, transforming corporate workforce capabilities across client organizations.
- **Core Skills & Technical Environment:** ForMinds , Chief Technology Officer (CTO) , Startup Co-Founder , Pharmaceutical Demand Forecasting , Quantitative Modeling , Time Series Forecasting , High-Precision Analytics , Enterprise AI Consulting , Corporate IT Training , Software Architecture , FastAPI , Python , Microservices , Docker , Cloud Security & Data Governance , GxP / GDPR Compliance , Executive Leadership .

Lead AI & Machine Learning Instructor | Enterprise Talent Evaluator | EY (Ernst & Young)

Aug 2025 – Oct 2025 & Jul 2026 • Italy (National Headquarters & EY Bari Hub / Hybrid)

- **National Enterprise AI Academy (Aug – Oct 2025):** Spearheaded and delivered an intensive, high-level AI & Software Engineering Academy across **EY Italy** at a national scale, upskilling engineering talent on cutting-edge production AI stacks.
- **Agentic AI & Multi-Agent Orchestration:** Designed comprehensive curriculums covering **Agentic AI**, multi-agent role-playing systems with **CrewAI**, complex LLM agent workflows using **LangChain / LangGraph**, and visual workflow orchestration with **n8n** for enterprise process automation.
- **LLM Evaluation, Benchmarking & MLOps:** Instructed rigorous AI evaluation frameworks and MLOps tooling, leveraging **DeepEval** and **MLflow** for hallucination detection, unit testing of LLM outputs, RAG evaluation metrics (Faithfulness, Answer Relevance), and model latency/cost tracking.
- **Talent Assessment & Real-World Capstone Project:** Led and supervised the end-to-end production of a real-world enterprise capstone project; acted as primary **technical evaluator for EY leadership**, directly assessing candidate code quality, architectural decisions, and performance to drive EY's national hiring selections.
- **Advanced ML & Data Analytics for Development Teams (Jul 2026 - EY Bari):** Delivered an advanced technical bootcamp for **EY Bari Hub's** junior and mid-level software developers, focusing on production **Machine Learning (ML)** pipelines, advanced data manipulation with **Python (Pandas, NumPy, Scikit-Learn)**, statistical feature engineering, and relational/vector database integration.
- **Core Skills & Technical Environment:** EY (Ernst & Young) , Agentic AI , Multi-Agent Systems , CrewAI , LangChain , n8n (Workflow Automation) , DeepEval (LLM Evaluation) , MLflow (MLOps) , Artificial Intelligence (AI) , Machine Learning (ML) , Deep Learning (DL) , Large Language Models (LLMs) , RAG Evaluation , Python , Scikit-Learn , Pandas , NumPy , Vector Databases , Relational Databases (SQL) , Technical Talent Assessment , Hiring & Recruitment Evaluation , Big 4 Enterprise Consulting .

Senior Generative AI & Digital Media Consultant | Triboo Group

Apr 2026 (1 mo) • Milan, Italy (On-Site)

Audience: Subsidiary Agencies & Publishing Divisions across the entire Triboo Group

- **Group-Wide Enterprise AI Transformation:** Delivered an intensive, one-month on-site AI for Marketing & Digital Media masterclass in Milan, training editorial, marketing, and agency teams across all subsidiaries of **Triboo Group**.
- **Google Gemini Suite & Multimodal Ecosystem:** Spearheaded deep-dive implementation of the **Google Gemini Suite** (Gemini Enterprise, Google Cloud Vertex AI, Google Workspace AI), leveraging advanced multimodal capabilities for marketing asset creation and operational efficiency.
- **AI-Assisted Journalism & Editorial Automation:** Architected automated newsroom workflows for digital journalism, real-time article outlining, SEO-optimized copy synthesis, source verification, and structured publishing pipelines.
- **Multilingual Localization & Cross-Border Publishing:** Engineered automated translation and cultural adaptation pipelines using large language models, accelerating international distribution of digital content without loss of tone or stylistic nuance.
- **Generative Visual Synthesis for Advertising & E-Commerce:** Guided creative teams in deploying generative image workflows for advertising campaigns, social media collateral, e-commerce merchandising, and multimodal visual experimentation.
- **Core Skills & Technical Environment:** Triboo Group , Google Gemini Suite , Google Cloud Vertex AI , Google Workspace AI , Generative AI for Marketing , AI for Journalism & Publishing , Multimodal AI , Automated Translation & Localization , Generative Copywriting , SEO Optimization , Image Generation , Prompt Engineering , Digital Publishing , Digital Transformation , Milan .

Senior Technical Instructor & AI Consultant | ESA (European Space Agency) (via Corporate Training Programs)

2026 • Italy / Europe (Remote & On-Site)

- **Vertical Technical Upskilling for Space Sector Personnel:** Designed and delivered specialized technical training and process modernization courses for **ESA (European Space Agency)** teams, focusing on cloud-native tooling, data manipulation, and modern generative AI workflows.
- **Prompt Engineering for Office & Process Automation:** Formulated advanced prompt engineering architectures tailored for document intelligence, technical report synthesis, administrative office automation, and inter-departmental communication efficiency.
- **Containerization & Dev Tooling (Docker):** Instructed hands-on modules on **Docker** containerization, Dockerfile optimization, multi-stage image builds, environment parity, and reproducible local/server execution.
- **Relational Data Querying & SQL Optimization:** Lectured practical **SQL** tracks covering relational data extraction, complex joins, subqueries, and structured data handling for operational workflows.
- **AI for Business Strategy & Decision Support:** Delivered executive sessions on **AI for Business**, evaluating generative AI tooling integration, process re-engineering, digital transformation roadmaps, and measurable productivity ROI.
- **Core Skills & Technical Environment:** European Space Agency (ESA) , Docker , Containerization , SQL , Relational Databases , Prompt Engineering , Office Automation , AI for Business , Workflow Automation , Generative AI , Process Optimization , Digital Transformation , Technical Upskilling , Corporate Training .

Senior Enterprise AI Consultant & Prompt Engineering Specialist | Enterprise Corporate Advisory

2024 – Present • Italy (Hybrid & On-Site for Enterprise Clients)

Key Enterprise Client Portfolio: Banca Monte dei Paschi di Siena (MPS), UniCredit, Autostrade per l'Italia, Piaggio Group, Lear Corporation (Fortune 500)**

- **Banking & Fintech AI Integration (UniCredit & Monte dei Paschi di Siena):** Architected production-grade prompt engineering frameworks, deterministic JSON schema outputs, and automated LLM workflow pipelines for corporate banking, ensuring strict compliance with financial privacy regulations, audit standards, and data governance.
- **Critical Infrastructure Workflow Optimization (Autostrade per l'Italia):** Engineered custom AI workflow integrations embedding LLMs into internal ticketing, incident reporting, and operational documentation pipelines, substantially reducing manual cycle times and operational friction.
- **Generative Visual AI & Industrial Design Synthesis (Piaggio Group):** Spearheaded cutting-edge **Generative Image & Multimodal AI** consulting (leveraging deep technical background in professional photography and video production). Deployed workflows using **Stable Diffusion (SDXL / ComfyUI / ControlNet)**, **Midjourney**, and **DALL-E 3** to accelerate product design prototyping, digital asset creation, and marketing visualization.
- **Global Manufacturing Process Re-engineering (Lear Corporation - Fortune 500):** Delivered strategic AI workflow integration and prompt architecture consulting for corporate engineering teams, optimizing supply-chain communication, internal knowledge extraction, and cross-departmental documentation.
- **End-to-End Enterprise Workflow Embeddings:** Authored robust system prompts, Chain-of-Thought (CoT) reasoning templates, and automated API-driven middleware (Python, REST, n8n) to seamlessly integrate AI intelligence into existing enterprise ERP, CRM, and collaboration ecosystems.
- **Core Skills & Technical Environment:** Prompt Engineering , System Prompting , Few-Shot Learning , Chain-of-Thought (CoT) , Generative AI , Image Generation (Generative Vision) , Stable Diffusion (SDXL) , ComfyUI , ControlNet , Midjourney , DALL-E 3 , Multimodal AI , Workflow Automation , n8n , Enterprise AI Integration , Banca Monte dei Paschi di Siena (MPS) , UniCredit Group , Autostrade per l'Italia , Piaggio Group , Lear Corporation (Fortune 500) , Banking AI Compliance , Data Privacy & GDPR , Digital Transformation , Python , REST APIs .

Senior Data Analytics & Applied AI Corporate Lecturer & Consultant | Profice

2025 – Present • Italy (Remote & On-site Corporate Client Programs)

- **Dual-Track Corporate AI & Analytics Programs:** Designed and delivered specialized corporate training and consulting initiatives tailored for two distinct audiences across diverse enterprise clients: **Technical Engineering Teams** and **Executive / C-Level Management**.
- **Technical Track (Engineers & Data Specialists):** Instructed deep-dive technical curriculums on **Advanced Data Analytics**, exploratory data analysis (EDA), predictive statistical modeling, supervised/unsupervised **Machine Learning (ML)** pipelines (**Pandas, NumPy, Scikit-Learn, SciPy, Statsmodels**), and advanced **SQL** (Window Functions, CTEs).
- **Applied AI & LLM Systems for Developers:** Spearheaded hands-on architectural workshops on **Generative AI** engineering, covering **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)** systems, vector embeddings, and **Vector Databases (ChromaDB, FAISS, LangChain)** for intelligent enterprise search and workflow automation.
- **Executive & Management Track (Directors & Decision-Makers):** Led strategic **Applied AI for Management** masterclasses, translating cutting-edge AI capabilities into actionable business strategies, ROI frameworks, workflow automation opportunities, risk assessment, and ethical **AI Governance**.
- **Enterprise Client Advisory:** Partnered with multiple enterprise client accounts across different business verticals, aligning curriculum design to specific corporate datasets, operational pain points, and strategic digital transformation objectives.
- **Core Skills & Technical Environment:** Data Analytics , Applied Artificial Intelligence (AI) , AI Strategy for Management , Executive Coaching , Machine Learning (ML) , Generative AI , Large Language Models (LLMs) , Retrieval-Augmented Generation (RAG) , Vector Databases (ChromaDB / FAISS) , Prompt Engineering , LangChain , Python , Pandas , NumPy , Scikit-Learn , SciPy , Statsmodels , Advanced SQL , Data Visualization (Plotly / Seaborn) , Business Intelligence (BI) , Corporate Digital Transformation , AI Governance & ROI , Multi-Client Consulting , Profice .

Senior Technical Trainer & Software Engineering Lecturer | ITConsulting (Forma.Temp Programs)

2024 – Present • Italy (Remote / Hybrid)

- **Enterprise Curriculum Engineering:** Designed and delivered comprehensive, production-oriented technical bootcamps and corporate training programs funded by **Forma.Temp**, accelerating candidate upskilling from fundamental software engineering to advanced enterprise paradigms.
- **Python, AI & Data Engineering Track:** Instructed end-to-end Python curriculums spanning Object-Oriented Programming (OOP), algorithmic complexity, relational data persistence (**SQL**), containerization (**Docker**), and applied **Artificial Intelligence (AI)**, covering **Machine Learning (ML)** algorithms, **Deep Learning (DL)** neural architectures, and data processing libraries (**Pandas, NumPy, Scikit-Learn, PyTorch**).
- **Enterprise Java & Microservices Track:** Lectured intensive backend modules centered on **Java (17/21)**, the **Spring Boot** ecosystem, **Spring Data JPA, Hibernate**, RESTful web API architecture, transaction management, and automated testing methodologies.
- **Modern Frontend & Immersive Creative Web:** Led advanced frontend masterclasses focusing on **React.js** component architecture, state management hooks, alongside cutting-edge creative development utilizing **Three.js** (WebGL 3D graphics rendering, shaders, canvas scenes) and **GSAP (GreenSock Animation Platform)** for high-performance interactive animations.
- **Production Standards & Code Mentoring:** Conducted rigorous architectural code reviews and guided students through real-world capstone projects, enforcing **Clean Code, SOLID principles**, Git branching strategies, and containerized deployment workflows with **Docker**.
- **Core Skills & Technical Environment:** Python , Artificial Intelligence (AI) , Machine Learning (ML) , Deep Learning (DL) , PyTorch , TensorFlow , Scikit-Learn , Pandas , NumPy , Docker , SQL , Relational Databases (MySQL/PostgreSQL) , Java , Spring Boot , Spring Data , RESTful APIs , React.js , Three.js , WebGL , 3D Graphics Rendering , GSAP (GreenSock) , JavaScript (ES6+) , HTML5 , CSS3 , Git / GitHub , Clean Code , SOLID Principles , Technical Training , Curriculum Design , Code Review , Forma.Temp , ITConsulting .

Senior Lecturer & Master Faculty (Advanced AI, CV, NLP & Full-Stack AI) | Epicode

2024 – Present • Italy (Remote)

Program: Master in Python, AI & Machine Learning (Advanced Specialization Tracks & Capstone)

- **Deep Learning Specialization:** Instructed the advanced modules of the Master program covering deep neural networks, loss functions, backpropagation mechanics, convolutional neural networks (**CNNs**), recurrent models, and modern **Transformer** architectures utilizing **PyTorch** and **TensorFlow**.
- **Computer Vision (CV) & Natural Language Processing (NLP):** Led intensive modules on real-time computer vision, object detection, and image segmentation (**OpenCV**, **YOLO**), alongside modern NLP pipelines, text embeddings, semantic tokenization, and transformer models via **Hugging Face**.
- **Generative AI & Full-Stack AI Process:** Directed end-to-end full-stack AI deployment, teaching students how to bridge the gap between model inference and production apps via **FastAPI**, **Docker**, microservice APIs, Retrieval-Augmented Generation (**RAG**), and frontend integration.
- **Capstone Project Supervision & Final Evaluation:** Acted as technical supervisor and lead committee evaluator for final real-world **Capstone Projects**, auditing software architecture, model accuracy, engineering best practices, and code maintainability.
- **Core Skills & Technical Environment:** Epicode , Master in Python, AI & Machine Learning , Deep Learning (DL) , Computer Vision (CV) , Natural Language Processing (NLP) , OpenCV , YOLO (Object Detection) , Transformers , Hugging Face , PyTorch , TensorFlow , Generative AI , Full-Stack AI , FastAPI , Model Serving & APIs , Docker , Python , Capstone Project Mentorship , Technical Assessment , Neural Networks (CNNs/RNNs) .

Senior Cloud ML Instructor & AWS Specialist | Sinervis Consulting (Client: Argo 3000)

Jan 2026 – Mar 2026 (3 mos) • Italy (Corporate Delivery)

Curriculum: AWS Machine Learning Engineer (Intermediate) for Enterprise Technical Teams

- **Comprehensive AWS ML Curriculum Delivery:** Instructed the full **AWS Machine Learning Engineer (Intermediate)** technical curriculum for the core software and engineering staff of **Argo 3000**, operationalizing production cloud ML practices.
- **Amazon SageMaker & MLOps Pipelines:** Directed practical implementation of **Amazon SageMaker** ecosystems (SageMaker Studio, Pipelines, Model Registry, Feature Store), automated hyperparameter tuning (HPO), and distributed training using custom containerized images.
- **Cloud Data Ingestion & Serverless ETL:** Guided teams through scalable cloud data engineering patterns leveraging **Amazon S3**, **AWS Glue**, **Amazon Athena**, and event-driven data pipelines triggered via **AWS Lambda**.
- **Zero-Downtime Model Serving & Observability:** Taught deployment strategies for production SageMaker endpoints (Real-Time, Serverless, Multi-Model Endpoints), auto-scaling policies, canary testing, and **SageMaker Model Monitor** for detecting data and concept drift.
- **Enterprise Cloud Security & Cost Optimization:** Implemented architectural security standards for cloud AI, including IAM role governance, KMS data encryption, VPC private endpoints for data isolation, and spot instance training for cost reduction.
- **Core Skills & Technical Environment:** Amazon Web Services (AWS) , AWS Machine Learning Engineer , Amazon SageMaker , SageMaker Pipelines , SageMaker Feature Store , SageMaker Model Registry , SageMaker Model Monitor , AWS Lambda , Amazon S3 , AWS Glue , Amazon Athena , MLOps on AWS , Model Serving & Endpoints , Hyperparameter Optimization (HPO) , Docker on AWS , Data & Concept Drift , VPC & IAM Security , Sinervis Consulting , Argo 3000 , Cloud Architecture .

Founder, Lead Software Architect & Data Scientist | Ultralimes Marketing

2025 – Present • Pisa / Remote, Italy

(Full R&D & Engine Development in 2025 • Commercial Rollout & Live Production Deployment in 2026)

Key Enterprise & Commercial Clients: Profice, ACUS, Tuscany Likes Events and PR**

- **End-to-End Product Conception & SaaS Engineering:** Conceived, architected, and engineered a proprietary AI-powered market intelligence and marketing analytics SaaS platform from scratch through 12 months of dedicated R&D in 2025, reaching full commercial rollout in 2026.
- **Custom Enterprise Solutions & Multi-Property PMS (Tuscany Likes Events and PR):** Engineered a comprehensive multi-villa luxury hospitality platform and custom Property Management System (PMS); unified booking calendars, real-time room availability, lease management, and reservation workflows across an entire portfolio of high-end luxury villas in Pisa.
- **Cloud Infrastructure & Server Administration (DevOps):** Configured, deployed, and maintain production Linux cloud environments (Ubuntu, Docker containers, Nginx reverse proxy, SSL/TLS, automated CI/CD pipelines, firewall security, and automated database backups).
- **High-Performance Backend & Asynchronous APIs:** Engineered resilient RESTful APIs and microservices using **Python (FastAPI)** and asynchronous runtimes, backed by **PostgreSQL** relational storage and **Redis** caching layers to ensure low-latency data querying under concurrent traffic.
- **Distributed Web Scraping & Data Extraction Pipelines:** Designed large-scale, anti-bot-resistant data harvesting pipelines leveraging **Playwright**, **BeautifulSoup**, and asynchronous task workers (**Celery** / **asyncio**), extracting, cleaning, and structuring millions of web data points.
- **Proprietary Predictive Algorithms & AI Integration:** Developed custom statistical models and predictive market analysis algorithms; integrated LLM synthesis pipelines to transform unstructured web data into automated sentiment reports and tailored commercial intelligence.
- **Commercial Growth & Team Management:** Sourced and negotiated enterprise accounts (including **Profice** and **ACUS**); manage and mentor a part-time technical collaborator, overseeing task delegation, code quality audits, and operational deliverables.
- **Core Skills & Technical Environment:** SaaS Architecture , Multi-Tenant Architecture , Property Management Systems (PMS) , Hospitality Tech & Booking Engines , Python , FastAPI , PostgreSQL , Redis , Distributed Web Scraping , Playwright , Selenium , BeautifulSoup , Asyncio , Celery (Task Queues) , Proprietary Predictive Algorithms , Market Intelligence , Docker , Nginx , Linux Server Administration (DevOps) , CI/CD Pipelines , RESTful APIs , Tuscany Likes Events and PR , ACUS , Profice , B2B Client Acquisition , Team Leadership , Startup Founder , Ultralimes Marketing .

Co-Owner & Head of Video Production | Golden Bloom Studio

2023 – Present • Tuscany / Pisa, Italy

(Co-Owned with Martina Brui - Head of Photography)

Key Enterprise & High-Profile Accounts: Cantieri Navali Codecasa (200th Anniversary Celebration), Solvay Group (Massa Plant)**

- **Creative Leadership & Video Department Management:** Direct technical and creative operations for the video department of **Golden Bloom Studio**, partnering with Co-Owner Martina Brui (Head of Photography) to deliver cinematic visual productions for luxury brands, corporate giants, and destination weddings across Tuscany.
- **Bicentennial Luxury Yacht Production (Cantieri Navali Codecasa):** Directed and produced the official celebration film for the prestigious **200-Year Bicentennial of Cantieri Navali Codecasa** (one of the world's most renowned luxury mega-yacht shipbuilders), capturing maritime engineering excellence, gala events, and legacy heritage.
- **Industrial Media Production (Solvay Group):** Produced high-impact corporate, technical, and industrial media assets for the **Solvay Massa Plant**, adhering to strict industrial safety standards and corporate visual guidelines for a global multinational.
- **End-to-End Post-Production & Color Grading:** Mastered high-end post-production pipelines using **DaVinci Resolve**, **Adobe Premiere Pro**, and **After Effects**; executed advanced color grading (ACES / RAW pipelines, LUT development, dynamic range mastering) and multi-track audio engineering.
- **Synergy with Ultralimes Marketing Engine:** Engineered seamless multimedia content pipelines providing high-resolution photo and cinematic video assets for digital marketing funnels, luxury hospitality web portals (such as the Tuscany Likes Multi-Villa PMS), and multimodal visual AI pipelines.
- **Core Skills & Technical Environment:**

Digital Cinema & Video Production , Cinematography , Video Editing , Color Grading , DaVinci Resolve , Adobe Premiere Pro , Adobe After Effects , ACES Color Management , RAW Video Pipelines , Lighting Design , Audio Post-Production , Creative Direction , Cantieri Navali Codecasa (200 Years Anniversary) , Solvay Group (Massa) , Luxury Destination Weddings & Events , Multimedia Marketing , Cross-Platform Synergy with Ultralimes , Golden Bloom Studio .

COMPREHENSIVE TECHNICAL SKILL MATRIX (ATS & ALGORITHMIC INDEX)

1. Artificial Intelligence, LLMs & Agentic Systems

- **Agentic AI & Multi-Agent Orchestration:** Agentic AI Architectures, Multi-Agent Systems, CrewAI (Role-Playing Autonomous Agents), LangChain, LangGraph, Autonomous Tool Calling & Execution, n8n Workflow Automation.
- **Large Language Models (LLMs) & Generative AI:** OpenAI GPT-4 / GPT-4o, Google Gemini Suite (Gemini 1.5 Pro/Flash, Gemini Advanced, Gemini Enterprise), Anthropic Claude, Open-Source LLMs (Llama 3, Mistral, Qwen).
- **Prompt Engineering & Cognitive Architectures:** Advanced System Prompting, Chain-of-Thought (CoT), Tree-of-Thoughts, Few-Shot / Zero-Shot Prompting, ReAct Framework, Structured JSON / Pydantic Output Parsing, Context Window Optimization.
- **Retrieval-Augmented Generation (RAG) & Vector Search:** RAG Architectures, Vector Embeddings, Semantic Search, Chunking Strategies, Vector Databases (**ChromaDB, FAISS, Pinecone, Qdrant**).
- **LLM Evaluation & MLOps: DeepEval, MLflow,** LLM Output Benchmarking, Hallucination Detection & Mitigation, RAG Triad Metrics (Context Relevance, Faithfulness, Answer Relevance), Model Observability, Token Cost & Latency Optimization.

2. Data Science, Machine Learning & Deep Learning

- **Deep Learning & Neural Architectures:** **PyTorch**, **TensorFlow**, Keras, Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs, LSTMs), Transformer Architectures, Backpropagation & Optimization Algorithms.
- **Computer Vision (CV):** **OpenCV**, **YOLO** (Real-Time Object Detection & Tracking), Image Classification, Semantic Segmentation, Feature Matching.
- **Natural Language Processing (NLP):** **Hugging Face Transformers**, Pretrained Tokenizers, Sentence Transformers, Spacy, NLTK, Sentiment Analysis, Text Embeddings.
- **Statistical Modeling & Quantitative Analytics:** Probability Theory, Mathematical Statistics, Regression Analysis (Linear, Logistic, Polynomial), Hypothesis Testing, Feature Engineering, Exploratory Data Analysis (EDA), Dimensionality Reduction (PCA).
- **Time Series & Predictive Analytics:** Pharmaceutical Demand Forecasting, Market Intelligence Modeling, Predictive Algorithms, Time Series Analysis (**ARIMA**, **Prophet**, **LightGBM**, **XGBoost**, **DeepAR**).
- **Scientific Computing:** **Pandas**, **NumPy**, **SciPy**, **Statsmodels**, **Matplotlib**, **Seaborn**, **Plotly**.

3. Generative Visual AI, Cinematography & Creative Technologies

- **Generative Image & Multimodal Systems:** **Stable Diffusion (SDXL, SD 1.5)**, **ComfyUI** (Node-Based Visual Workflows), **ControlNet**, LoRA Fine-Tuning, **Midjourney**, **DALL-E 3**, Flux, Prompt-to-Image / Image-to-Image Pipelines, Multimodal Generation.
- **Video Editing & Digital Post-Production:** **DaVinci Resolve Studio**, **Adobe Premiere Pro**, **Adobe After Effects**, Final Cut Pro, High-Bitrate ProRes/RAW Media Pipelines.
- **Color Science & Audio:** ACES Color Management, Log to Rec.709 Color Science, Custom 3D LUT Engineering, HDR Mastering, Dynamic Range Optimization, Audio Post-Production & Sound Design.
- **Cinematography & Directing:** Digital Cinema Cameras (4K/6K), Lighting Design, Multi-Camera Live Direction, Corporate Documentary Production, Luxury Wedding Cinematography.

4. Backend Engineering, Microservices & Software Architecture

- **Core Programming Languages:** **Python** (Advanced / Asyncio), **Java** (Java 8, Java 11, Java 17, Java 21), **SQL**, **JavaScript (ES6+)**, **TypeScript**, **R** (Statistical Computing).
- **Backend Frameworks & Runtimes:** **FastAPI**, **Flask**, **Spring Boot 3**, **Spring Framework**, **Spring Data JPA**, **Hibernate ORM**, Celery (Distributed Task Queue), Asyncio.
- **Architecture & Design:** Microservices Architecture, Clean Architecture, Domain-Driven Design (DDD), Event-Driven Architecture (EDA), SOLID Principles, Design Patterns, RESTful API Design, OpenAPI / Swagger Documentation.
- **SaaS & Enterprise Systems:** Multi-Tenant SaaS Platforms, Property Management Systems (PMS), Booking & Reservation Engines, High-Concurrency Request Handling.

5. Frontend & Interactive Creative Web

- **Frontend Frameworks & Libraries:** **React.js** (Hooks, Context, State Management), Modern JavaScript (ES6+), HTML5, CSS3, Responsive Design.
- **3D & Interactive Web:** **Three.js**, **WebGL** (3D Canvas Scenes & Shaders), **GSAP (GreenSock Animation Platform)**, High-Performance Timeline Animations, Interactive UI/UX.

6. Databases, Storage, Caching & Data Extraction

- **Relational Databases (RDBMS):** **PostgreSQL**, **MariaDB**, **MySQL**, Oracle Database, Advanced SQL (Window Functions, Common Table Expressions - CTEs, Index Optimization).
- **Caching & In-Memory Stores:** **Redis** (Data Caching, Rate Limiting, Message Brokering).
- **Database Migrations:** Flyway, Schema Evolution, DDL/DML Optimization.
- **Distributed Web Scraping & Data Harvesting:** **Playwright**, **Selenium**, **BeautifulSoup4**, Anti-Bot Bypass, Dynamic Headless Crawling, Automated ETL Extraction Pipelines.

7. Cloud, DevOps, MLOps & Infrastructure

- **Cloud Platforms & AI Engines:** **Amazon Web Services (AWS)** (SageMaker Studio, Pipelines, Feature Store, Model Registry, S3, Lambda, Glue, Athena, IAM, VPC), **Google Cloud Platform (GCP)** (Vertex AI, Gemini Enterprise, Workspace AI).
- **Containerization & Deployment:** **Docker**, **Docker Compose**, Containerized ML Model Serving, Microservice Containers.
- **Server Administration & Networking:** **Linux Server Administration** (Ubuntu Server), **Nginx** (Reverse Proxy, SSL/TLS Termination, Load Balancing), VPS Management, SSH, Firewall Security (UFW).
- **CI/CD & Version Control:** **Git**, **GitHub**, **GitLab CI/CD**, **Azure DevOps**, Postman API Testing.

8. Enterprise Domains, Leadership & Governance

- **Executive Leadership:** Chief Technology Officer (CTO), Startup Co-Founder, Technical Talent Assessment & Hiring Panel Evaluation, Engineering Management, Team Mentoring & Code Review.
- **Corporate Client Portfolio:** European Space Agency (ESA), EY (Ernst & Young), Triboo Group, Banca Monte dei Paschi di Siena (MPS), UniCredit, Autostrade per l'Italia, Piaggio Group, Lear Corporation (Fortune 500), Cantieri Navali Codecasa, Solvay Group, Argo 3000, Epicode, Profice, ITConsulting.
- **Regulated Industries & Compliance:** Pharmaceutical Demand Forecasting (GxP & GDPR), Banking & Financial Compliance, AI Safety Guardrails, Data Privacy & Confidentiality.